

What Can Repeated Size Profiles Identify? Bayesian Inference and Confounding in Inverse Integral Projection Models

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Abstract

Repeated cross-sectional size profiles, meaning counts of individuals by body size collected over several years, are among the most common and least expensive data in population ecology. It is tempting to invert a size-structured integral projection model (IPM) against such data and read off survival, growth, and reproduction. We ask a prior question instead: which vital rates are identified by repeated profiles, which are confounded, and what additional information resolves them. Using an individual-based generator distinct from the recovery model, a positive numerical-quadrature inverse IPM, and a fully Bayesian treatment, we find that size profiles reliably recover survival, the growth increment, and the recruit-size distribution, while recruitment intensity is only weakly identified, along a single ridge in parameter space. Two identifiability results explain this. A decomposition result shows that the total per-capita contribution of each size is identified but its split into survival and recruitment is not, wherever growth and recruit sizes overlap. An excitation result shows that near a stable size distribution, profiles constrain the asymptotic growth rate λ and the stable shape far more tightly than the vital rates that generate them. We confirm both numerically, calibrate the posterior approximation by simulation-based calibration, quantify how size-dependent detection corrupts recovery, and measure how much auxiliary individual-level data are needed to break the confounding. A 20-year brook trout capture-mark-recapture study illustrates the practical risk: a profile-only fit can look entirely plausible yet badly misestimate growth and the size pattern of survival, which only the individual data reveal. We therefore present the work as a diagnostic framework for what is identified, how to detect confounding, and how to design the data or priors needed for credible inference, rather than as a claim that profiles generally identify vital rates.

Keywords: integral projection models; identifiability; inverse problems; Bayesian hierarchical models; integrated population models; simulation-based calibration; size-structured populations.

1 Introduction

Size-structured populations are routinely monitored by repeated cross-sectional surveys that record the body sizes of sampled individuals. The resulting *size profiles*, histograms of size at successive census times, are inexpensive relative to individual marking, and they exist for many taxa over

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long periods. The integral projection model (Easterling et al., 2000; Ellner and Rees, 2006; Merow et al., 2014) gives a natural mechanistic description of how such a profile should evolve: a kernel built from size-dependent survival, growth, and reproduction maps this year’s size distribution to next year’s. A seemingly direct strategy is therefore to invert the IPM against repeated profiles and recover the vital-rate functions. This is an instance of a classic inverse problem for structured populations (Wood, 1994; Gross et al., 2002; Ghosh et al., 2012), and like most inverse problems it is ill-posed: many different vital-rate configurations can generate nearly identical profile dynamics.

We argue that the right question is not “can we fit an inverse IPM to size profiles” (we almost always can) but “what does the fit identify.” A model can reproduce observed profile dynamics accurately while estimating a population-effective transition operator that differs from the individual vital rates that generated the data, and it can report narrow posterior intervals that are an artifact of the prior rather than the data. We therefore treat identifiability, confounding, and prior sensitivity as objects of study in their own right, declare the estimands before fitting, and validate recovery against data generated by a process other than the recovery model.

This paper makes five contributions. We separate an individual-based biological generator from the recovery model, so that the inverse IPM is never validated only on data from the same IPM. We characterize, analytically and numerically, which vital rates repeated profiles identify, and the confounding ridge between recruitment intensity and its size gradient. We give a calibrated Bayesian workflow, including simulation-based calibration of the posterior approximation and a demonstration that size-dependent detection, if unmodeled, is read as demography. We quantify how much auxiliary individual data break the confounding, including robust prior borrowing and its transportability limits. Finally, we apply the framework to a long brook trout time series, benchmarking profile-only inference against capture–mark–recapture and integrated analyses. The methods and diagnostics are meant to carry over to other size- and stage-structured inverse problems.

► **[COAUTHOR INPUT NEEDED, biology/ecology]:** *One to two paragraphs of ecological and management motivation: why size profiles are collected in your systems, what management or conservation decisions rest on vital-rate estimates from them, and why distinguishing identified from confounded rates matters in practice. Cite the key applied literature for the focal taxa.*

The remainder of the paper declares the estimands and notation (Section 2), specifies the generator, observation process, recovery model, priors, inference, and diagnostics (Section 3), develops the identifiability analysis (Section 4), reports the simulation experiments (Section 5), presents the brook trout application (Section 6), and discusses scope and limitations (Section 7).

2 Estimands and Notation

We distinguish four inferential targets and report them separately throughout, since good performance on one does not imply recovery of another:

1. *Individual vital rates:* survival probability $s(x)$, mean and spread of the growth transition $g(x' | x)$, realized recruitment intensity $r(x)$, and the recruit-size distribution $b(x' | x)$, as functions of individual size x .
2. *Population-effective rates:* the same quantities after averaging over unobserved heterogeneity, movement, and detection, which is the most a size-only model can target.
3. *Kernel functionals:* the dominant eigenvalue λ (asymptotic population growth rate), the stable size distribution, and sensitivities.
4. *Prediction targets:* next-year and multi-year size profiles.

Throughout, x denotes size on a fixed mesh, Δx the mesh spacing, μ_t the latent population size-intensity at census t , and Y_{tj} the observed count in size bin j at census t .

3 Methods

3.1 Individual-based generator

The data generator operates on individuals, not on the recovery kernel, so that recovery is never tested only against self-generated data. At census t , individual i has size x_{it} , age, sex q_i , and optional persistent latent quality u_i . Survival is an individual Bernoulli event,

$$S_{it} \sim \text{Bernoulli}\{s(x_{it}, q_i, u_i, E_t, N_t)\}, \quad (1)$$

with E_t an environmental covariate and N_t population size. Conditional on survival, next size is drawn from a growth distribution,

$$x_{i,t+1} \mid S_{it} = 1 \sim \mathcal{N}\{m_g(x_{it}), \sigma_g^2\}, \quad (2)$$

and realized recruitment is generated separately as an individual count process, with recruit sizes drawn independently of parent size up to its mean,

$$R_{it} \sim \text{Poisson}\{r(x_{it}) \mathbb{I}(q_i = \text{F}) \cdot c\}, \quad x_{\text{recruit}} \sim \mathcal{N}\{m_b(x_{it}), \sigma_b^2\}. \quad (3)$$

Female reproductive output is scaled so that its population average matches the declared recruitment curve. Survival, growth, and reproduction are thus distinct stochastic events, and the simulated population does not evolve by deterministic application of the recovery IPM.

► **[COAUTHOR INPUT NEEDED, biology/ecology]:** *Confirm that this individual-level generator matches the biology of the focal systems (e.g. semelparity vs. iteroparity, sexual size dimorphism, density dependence, environmental drivers). Note any structurally important process the generator omits.*

3.2 Size-profile observation process

Identities are retained by the generator but discarded before profile-only analysis. At each census, individuals are independently detected with probability $q_t(x)$, optionally with measurement error, and grouped into equal-width bins. The general observation intensity is $\lambda_t(x) = e_t q_t(x) \mu_t(x)$ with effort e_t . Conditional on the initial profile, subsequent bin counts follow either a Poisson model $Y_{tj} \sim \text{Poisson}(M_{tj})$ or a Gamma–Poisson model with negative-binomial margin (mean M_{tj} , precision ϕ) that accommodates overdispersion, where M_{tj} is the model-implied expected count.

3.3 Positive inverse IPM

The recovery model propagates latent size intensity through a decomposed, strictly positive kernel,

$$\mu_{t+1}(x') = \int K(x', x) \mu_t(x) dx, \quad K(x', x) = P(x', x) + F(x', x), \quad (4)$$

$$P(x', x) = s(x) g(x' \mid x), \quad F(x', x) = r(x) b(x' \mid x), \quad (5)$$

with both conditional size densities normalized, $\int g(x' | x) dx' = \int b(x' | x) dx' = 1$. The column integrals of P and F are then exactly $s(x)$ and $r(x)$, which removes arbitrary scaling between recruitment intensity and recruit-size shape. This normalization does not by itself identify the split of K into P and F ; that split must be learned from temporal information, restrictions, priors, or auxiliary data. The vital-rate functions are parameterized at a centering size of 80 with $z(x) = (x - 80)/20$:

$$\text{logit } s(x) = \alpha_s + \beta_s z, \quad m_g(x) - x = \max\{0, \alpha_g + \beta_g z\}, \quad \sigma_g = e^{\eta_g}, \quad (6)$$

$$\text{log } r(x) = \alpha_r + \beta_r z, \quad m_b(x) = \alpha_b + \beta_b z, \quad \sigma_b = e^{\eta_b}, \quad (7)$$

giving ten demographic parameters; the Gamma–Poisson model adds $\log \phi$. The continuous operator is discretized on a regular mesh, $\boldsymbol{\mu}_{t+1} = \Delta x \mathbf{K} \boldsymbol{\mu}_t$, with conditional normal transition probabilities integrated over target bins rather than evaluated at bin centers, which otherwise inflates recruit-size variance at coarse bins, and renormalized over the retained mesh.

3.4 Priors

We use independent weakly informative normal priors on the transformed coefficients; representative induced 95% prior intervals are biologically diffuse but retain structure through the link functions, parametric curve forms, normalized transition distributions, and the non-negative growth constraint. The Gamma–Poisson precision receives $\log \phi \sim \mathcal{N}\{\log 50, 2^2\}$. External information, when used, is expressed as an explicit auxiliary study design rather than only as abstract prior standard deviations (Section 5.7).

3.5 Inference

Posterior computation uses maximum *a posteriori* optimization with Gaussian Laplace approximations for the broad grids, which are fast and resumable, audited against adaptive-Metropolis MCMC on representative datasets, and validated by simulation-based calibration (Cook et al., 2006; Talts et al., 2018). Non-positive-definite posterior Hessians are recorded as approximation failures and excluded from coverage summaries. For the finite-population state-space variant we use a particle marginal Metropolis–Hastings sampler (reported in supplementary material); the present paper’s claims rest on the deterministic projection likelihood and its audited approximation.

3.6 Identifiability diagnostics

We report, for every fit, the prior-to-posterior contraction $1 - \text{sd}_{\text{post}}/\text{sd}_{\text{prior}}$ per parameter, which separates data-informed from prior-dominated quantities; the posterior correlation between recruitment level α_r and slope β_r , the leading confounded pair; and the condition number and least-informed eigenvector of the observed information, which expose weakly identified directions even when no single parameter is flat.

4 Identifiability Analysis

Two results explain, and predict, the empirical recovery patterns reported below. Both concern the discretized one-step map $\mathbb{E}[\boldsymbol{\mu}_{t+1}] = \Delta x \mathbf{K} \boldsymbol{\mu}_t$ with per-capita kernel entries $K_{ij} = s_j g_{ij} + r_j b_{ij}$, where column j collects the contributions of source size j to all destination sizes i .

4.1 Decomposition identifiability

Proposition 1 (Total contribution is identified; its split is not, on the overlap). *The column sum $\Delta x \sum_i K_{ij} = s_j + r_j$ equals the total per-capita contribution of source size j , because g and b are normalized. Hence, wherever the kernel column $K_{\cdot j}$ is identified, the sum $s_j + r_j$ is identified. The decomposition into survival-growth $s_j g_{\cdot j}$ and recruitment $r_j b_{\cdot j}$ is identified on destination sizes where the growth density $g(\cdot | j)$ and the recruit-size density $b(\cdot | j)$ have disjoint support, and is confounded on destination sizes where they overlap.*

Argument. On a destination size i in $\text{supp}(g(\cdot | j)) \setminus \text{supp}(b(\cdot | j))$ the entry $K_{ij} = s_j g_{ij}$ involves only survival-growth, and symmetrically for recruitment; on the overlap $K_{ij} = s_j g_{ij} + r_j b_{ij}$ constrains only the sum of two positive terms, so (s_j, r_j) and the local density values trade off freely subject to nonnegativity and the normalization constraints. $\square$

This matches what we observe. Recruit sizes are small and overlap the sizes that small survivors grow into, so recruitment intensity is confounded with small-fish survival and growth, while survival of larger sizes, whose grown distribution avoids the recruit band, is identified, as is the recruit-size distribution itself. The clearest practical consequence is the recruitment level/slope ridge of Section 5.1 (Figure 1): with all other parameters fixed, (α_r, β_r) trade off along a near-flat direction with posterior correlation below -0.95 .

4.2 Excitation and the role of stationarity

Proposition 2 (Near stationarity, only λ and the stable shape are identified). *Suppose the latent profiles are proportional to a fixed shape, $\mu_t = c_t \mathbf{w}$ for all t . Then the expected data depend on the kernel only through $\Delta x \mathbf{K} \mathbf{w} = \lambda \mathbf{w}$, with $\lambda = c_{t+1}/c_t$. Any kernel $\mathbf{K}'$ sharing this dominant eigenpair produces identical expected profiles, so the dominant eigenvalue λ and the stable shape $\mathbf{w}$ are identified, while the remaining $m(m-1)$ degrees of freedom of $\mathbf{K}$, and hence the vital rates beyond $(\lambda, \mathbf{w})$, are not. Identifying more of $\mathbf{K}$ requires the population to occupy multiple linearly independent states (excitation: cohort pulses, environmental variation, or transient non-stationary dynamics).*

The proposition makes precise the intuition that a population observed near its stable size distribution carries little information about its generating operator, and it predicts that the kernel functional λ is recovered far better than the rates that produce it. Section 6.3 bears this out in the brook trout data, where the profile-only posterior spreads across biologically very different vital-rate regimes that share nearly the same λ .

5 Simulation Experiments

5.1 The recruitment confounding ridge

A large, low-noise data set generated from the recovery model itself (one long trajectory, full detection, about 6000 individuals) isolates structural identifiability from Monte Carlo noise. The profile likelihood is sharply peaked in most directions but nearly flat along a single ridge that trades recruitment level against recruitment slope (Figure 1): the posterior correlation between α_r and β_r is -0.96 , the curvature of the two-dimensional log-likelihood implies a ridge correlation of

0.98, and the observed-information condition number exceeds 7×10^3 . The profile data constrain the total kernel K and the influx into each size bin, but not the partition of that influx between survival-growth flux and recruitment flux, so a higher recruitment level with a flatter size gradient produces nearly the same profile dynamics as a lower level with a steeper gradient.

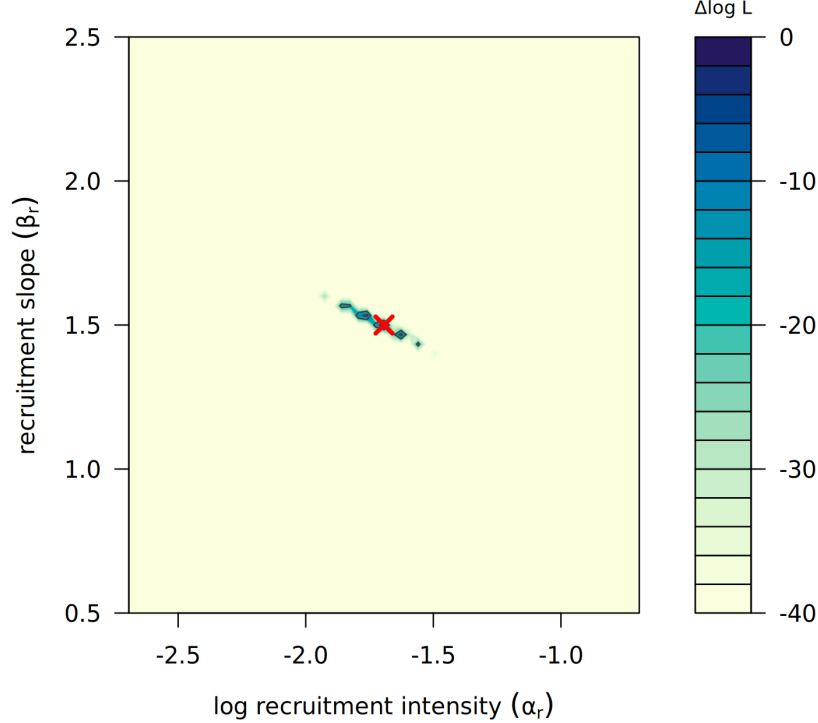


Figure 1: Relative profile log-likelihood over recruitment level α_r and slope β_r , all other parameters at truth (red cross), capped at $\Delta \log L = -40$ to reveal structure. The likelihood is sharply peaked except along a thin ridge oriented in the negative-correlation direction, which is the non-identified recruitment direction.

5.2 Excitation restores recruitment identifiability

Proposition 2 predicts that the confounding stems from near-stationarity and should weaken as the population is perturbed off its stable distribution. We tested this with matched data sets differing only in the initial state, interpolated from exactly the stable distribution $\mathbf{w}$ to a strong small-size cohort pulse, holding the vital rates, population size, and observation noise fixed (Figure 2). The dominant eigenvalue λ is recovered accurately at every excitation level (relative error below 0.01), including at exact stationarity, so the asymptotic functional is identified even when the rates are not. Survival remains identified throughout (contraction about 0.98). Recruitment intensity, by contrast, is nearly unrecoverable near stationarity (relative error 0.8 to 1.1, level/slope posterior correlation below -0.99) and is progressively recovered as excitation increases, reaching relative error 0.03 and level/slope correlation -0.31 under the strongest cohort pulse. The confounding is thus excitation-limited: a perturbed population, following a disturbance, a strong cohort, or environmental forcing, carries the temporal information needed to separate recruitment from survival-growth, whereas one observed near equilibrium does not. The practical implication is a design recommendation. To recover recruitment from size profiles, observe the population away from its stable distribution.

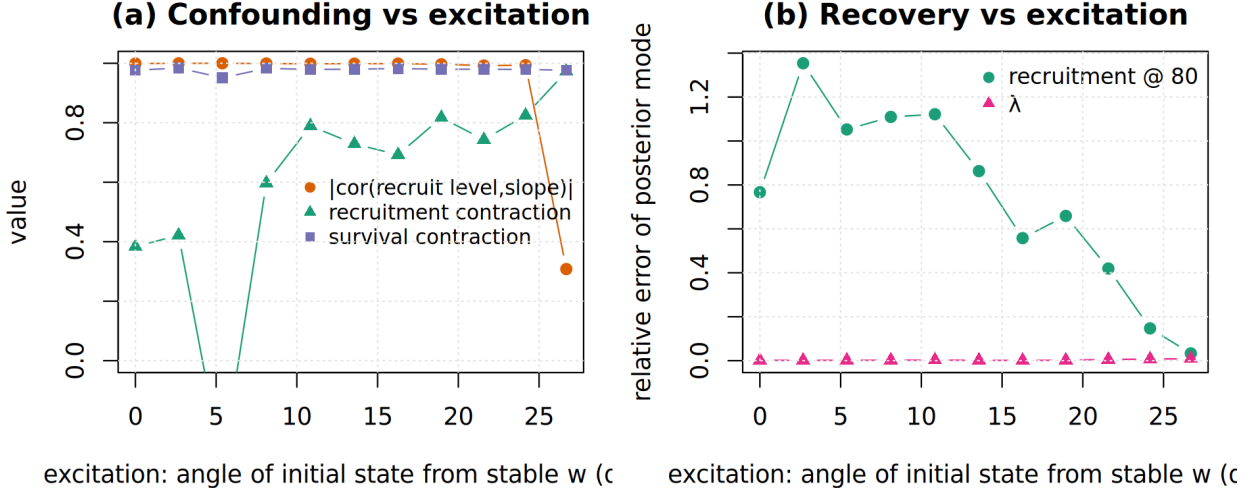


Figure 2: Excitation experiment. As the initial state is perturbed further from the stable distribution (x-axis: angle from $\mathbf{w}$): (a) the recruitment level/slope confounding breaks and recruitment contraction rises, while survival stays identified; (b) recruitment intensity recovery improves by an order of magnitude, while λ is recovered accurately at all excitation levels.

5.3 Robustness to flexible vital-rate forms

To check that the confounding is not an artifact of the parametric vital-rate forms, we refit data generated from the parametric truth with a flexible model in which survival, the growth increment, log-recruitment, and recruit mean are each natural-cubic-spline functions of size (19 parameters; weakly informative, lightly shrunk coefficient priors). The flexible model recovers survival, growth, and the recruit-size curve closely over the data-rich size range, but fails to recover recruitment intensity, which stays near zero and misses the true exponential rise (Figure 3). Averaged over sizes, the posterior coefficient of variation of the recruitment curve is 17.5 times that of survival (0.57 vs. 0.033). The confounding is therefore structural, the decomposition limit of Proposition 1, and not a consequence of the parametric kernel.

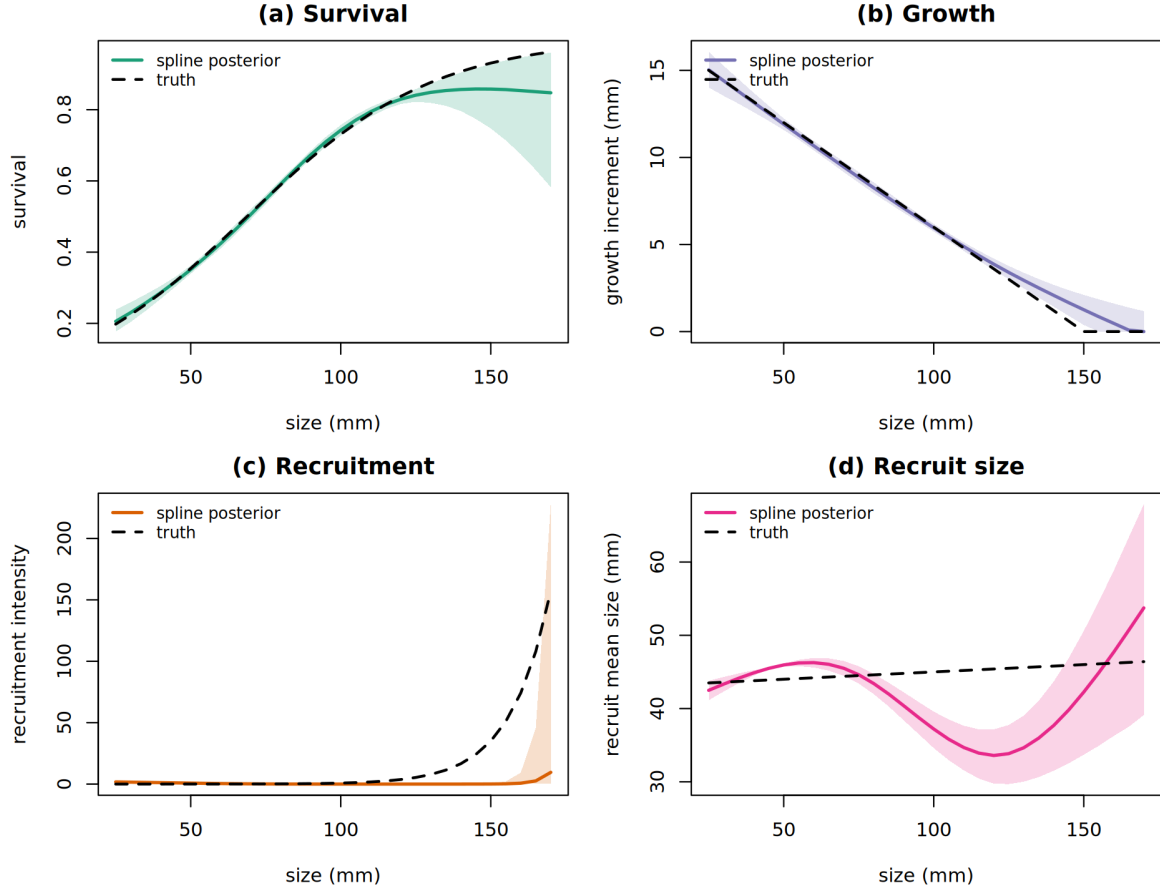


Figure 3: Flexible spline inverse IPM fitted to data from the parametric truth (dashed lines), with 95% posterior bands. Survival (a), growth (b), and recruit size (d) are recovered over the data-rich size range; recruitment intensity (c) is not, remaining near zero with a wide band. The confounding persists nonparametrically.

5.4 Broad recovery grid

A factorial Monte Carlo grid varied the allocation of a fixed 12-transition budget (1×12 , 3×4 , 6×2), initial population size, profile detection, initial profile shape, and recovery likelihood (Poisson vs. Gamma-Poisson), producing 240 simulated datasets and 480 fits. Replicated short trajectories outperformed one long trajectory on average (Table 1), but no design achieved nominal coverage for all quantities. Recovery was strongly quantity-dependent (Figure 4): the recruit-size distribution, survival, and mean growth were recovered well, whereas recruitment intensity and the dominant eigenvalue were biased and overconfident, the simulation counterpart of the ridge in Section 5.1.

Table 1: Mean 95% interval coverage by allocation of the 12-transition budget.

Design	Poisson params	Poisson derived	GP params	GP derived
1×12	0.734	0.701	0.764	0.718
3×4	0.795	0.775	0.818	0.818
6×2	0.795	0.813	0.816	0.836

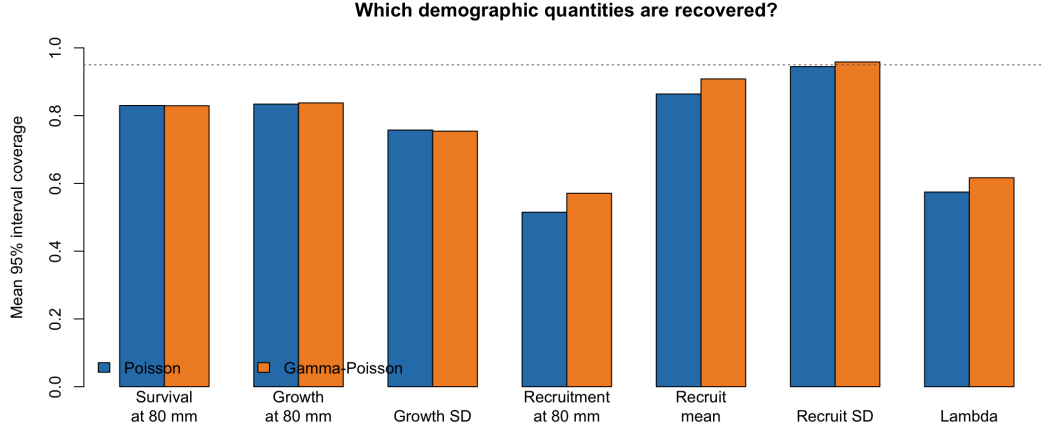


Figure 4: Interval coverage varied sharply among demographic quantities. Recruit size, survival, and mean growth were recovered more reliably than recruitment intensity and the dominant eigenvalue.

5.5 Calibration of the posterior approximation

We validated the Laplace posterior by simulation-based calibration, generating data from the recovery model’s own data-generating process so that rank non-uniformity isolates computational and approximation error rather than misspecification (Figure 5). Under an informative prior, 200 of 200 fits succeeded and 10 of 11 parameters had uniform rank statistics, with only the overdispersion parameter $\log \phi$ mildly miscalibrated. Under a diffuse prior the rank histograms became U-shaped, indicating overconfident intervals, most severely along the confounded recruitment and growth directions. The practical implication is that, under diffuse priors, interval inference for the confounded directions should come from full posterior sampling rather than the Gaussian approximation.

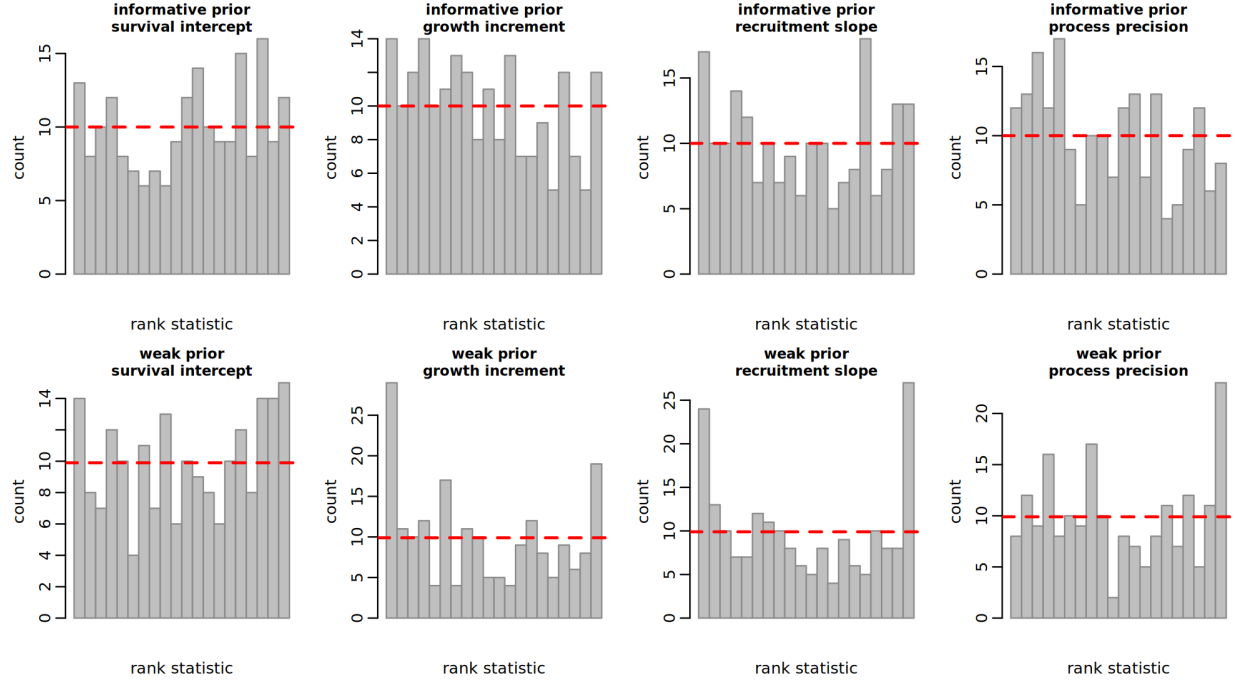


Figure 5: Simulation-based calibration rank histograms (dashed line: uniform expectation). Top: informative prior, ranks approximately uniform. Bottom: diffuse prior, U-shaped ranks indicating overconfident intervals, worst for the recruitment slope.

5.6 Observation-model robustness

A constant, size-independent detection probability is absorbed by the linear projection and does not bias the profile-only fit. Size-dependent detection does. We simulated three regimes (constant detection, unknown time-varying effort, and size-dependent detection) and fit the standard recovery model (Figure 6). Under constant detection, survival-at-80 coverage was 0.90–0.95; under time-varying effort it fell to about 0.65; under size-dependent detection it collapsed to zero, because a size-dependent change in detectability is absorbed as demographic change. This is the main practical threat to profile-only inference, and it motivates the caveats in the application.

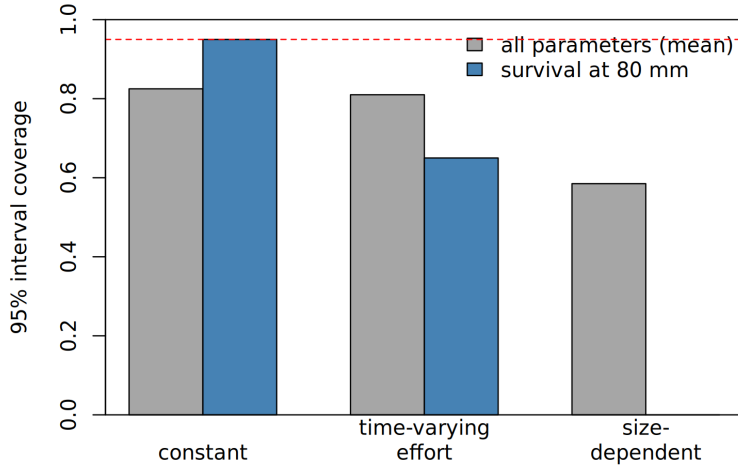


Figure 6: 95% interval coverage (Gamma–Poisson recovery) under three detection regimes. Size-dependent detection collapses survival coverage; the dashed line is the nominal 0.95 level.

5.7 Auxiliary data and transportability

Because recruitment intensity is the least identified rate, we quantified how much external individual-level fecundity information resolves it, expressed as an explicit auxiliary study of n_e fish analyzed by Poisson regression whose (approximate) posterior becomes the prior for (α_r, β_r) . Correctly transportable external data improved recruitment recovery monotonically in n_e ; 800 external fish cut recruitment-curve RMSE by about 70% and raised recruitment-at-80 coverage from 0.53 to 0.87. A precise but biased external study produced low point-estimate error yet zero coverage; robust (mixture) borrowing mitigated but did not eliminate the damage, and could reject a non-transportable study only when the conflict altered profile dynamics along a likelihood-informed direction (Figure 7). No single amount of external-study bias is universally safe.

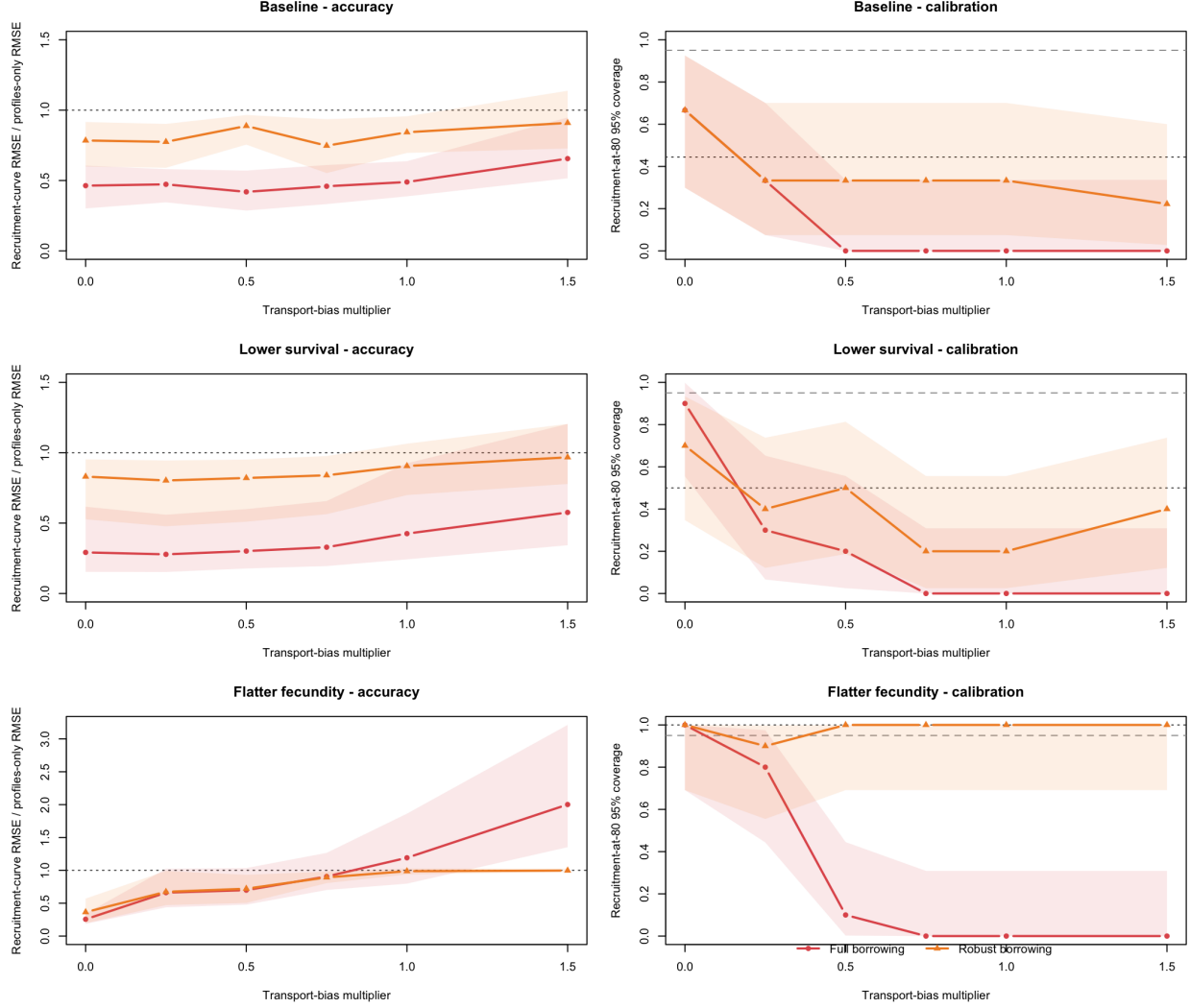


Figure 7: Accuracy and calibration as external-study bias increases (pooled across biological truths; 30 replicates per cell). Full borrowing improves point error but loses calibration at the smallest tested bias; robust borrowing degrades more gracefully. Dotted lines: profiles-only performance; dashed: nominal coverage.

5.8 Paired-data benchmark

On 200 datasets each carrying both profiles and capture–recapture histories, we compared profile-only, capture–recapture-only, integrated, and complete-history oracle analyses (Figure 8). Capture–recapture efficiently identified survival and growth but carried essentially no information about recruitment; its apparently perfect recruitment coverage reflected prior-width intervals, not recovery. Integration improved point accuracy for survival and growth but narrowed intervals below nominal, as residual discrepancy between observed and deterministically projected profiles was absorbed by other parameters, a point we return to in the application.

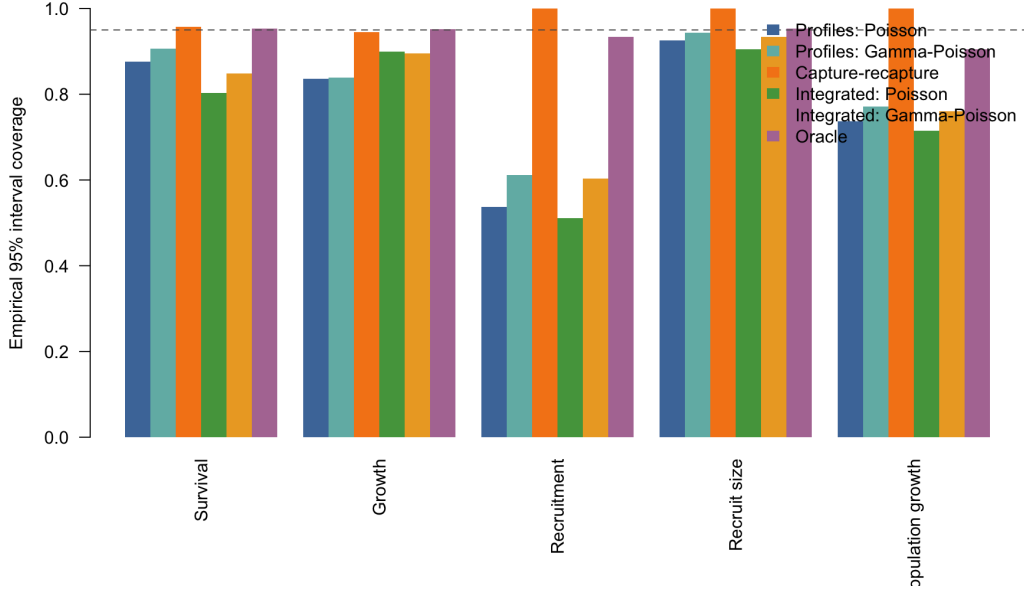


Figure 8: Empirical interval coverage in the paired-data benchmark. Perfect capture–recapture coverage for recruitment and population growth reflects prior-dominated wide intervals, not data-based identification.

6 Application: West Brook Brook Trout

6.1 Data and study system

We applied the framework to a long-term passive-integrated-transponder (PIT) tagging study of brook trout (*Salvelinus fontinalis*) in the West Brook, Massachusetts, USA (Letcher et al., 2015), obtained from the U.S. Geological Survey (U.S. Geological Survey, 2024). We treat the whole stream network (main stem and three tributaries) as one population, because a non-trivial fraction of fish move between reaches each year and a main-stem-only boundary would count emigrants as deaths. The profile stream is the standardized seasonal size cross-section over 2000–2019 (20 annual occasions, 19 transitions), restricted to sizes ≥ 70 mm where electrofishing detection is approximately size-uniform; recruitment is modeled as entry into this observable class. The size profiles are close to stationary across the two decades (Figure 9).

► [COAUTHOR INPUT NEEDED, biology/ecology]: *Expand this into a proper study-system description: stream characteristics, sampling design and electrofishing protocol, age/size structure of brook trout in this system, the biological meaning of “recruitment to the ≥ 70 mm class” here, and why the population may be near a stable size distribution. Add references to the field program.*

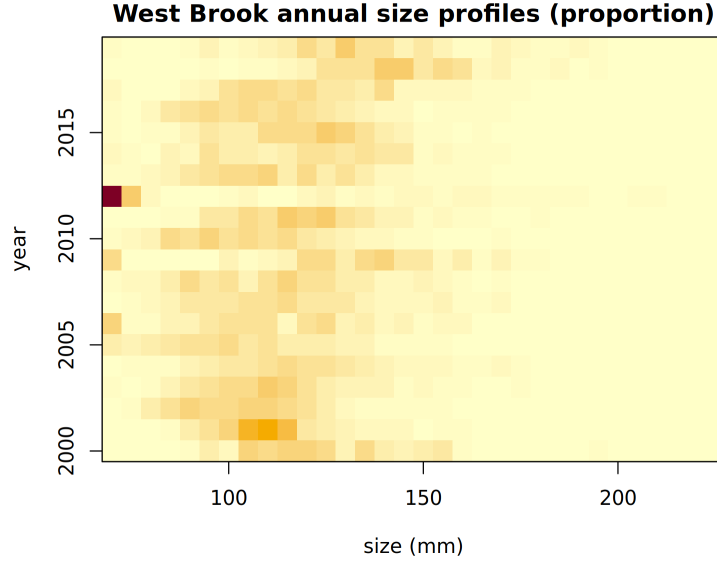


Figure 9: Observed annual size profiles (proportion by 5 mm bin) of West Brook brook trout, 2000–2019. The distribution is close to stationary across years.

6.2 Profile-only, mark–recapture, and integrated analyses

We fit the profile-only inverse IPM, a size-structured mark–recapture model for survival and growth (with detection estimated at $\hat{p} = 0.97$ from a constant-rate Cormack–Jolly–Seber model over the full histories, 20,077 recapture events), and an integrated model combining both likelihoods. Growth estimated from recaptures is detection-free and serves as the cleanest cross-check.

The profile-only fit is internally plausible (moderate survival, positive growth, $\lambda \approx 1$, a positive-definite posterior), yet it is contradicted by the individual data on both quantities the recaptures can check (Table 2, Figure 10). It underestimates the growth increment (12.6 vs. 32.3 mm yr^{−1}), the signature of a near-stationary size distribution that hides how fast individuals move through it, and it misses the size gradient in apparent survival (roughly flat near 0.44, against a clear decline from 0.31 to 0.12 with size in the recapture and integrated analyses, reflecting greater emigration of larger fish). Recruitment intensity remains unidentified by profiles (level/slope posterior correlation -0.995), as in the simulations. The integrated analysis adopts the recapture survival and growth and tightens the population growth rate to $\lambda = 1.016$ [1.00, 1.05].

Table 2: West Brook brook trout: posterior median [95% interval] by analysis. Mark–recapture survival is apparent survival, which folds in size-dependent emigration from the network.

Quantity	Profile-only	Mark–recapture	Integrated
Survival @ 80 mm	0.43 [0.29, 0.60]	0.31 [0.31, 0.32]	0.31 [0.31, 0.32]
Survival @ 110 mm	0.45 [0.35, 0.55]	0.20 [0.20, 0.21]	0.20 [0.19, 0.21]
Survival @ 140 mm	0.46 [0.34, 0.58]	0.12 [0.12, 0.13]	0.12 [0.11, 0.12]
Growth incr. @ 80 mm	12.6 [9.3, 15.6]	32.3 [31.7, 32.8]	32.1 [31.5, 32.7]
λ	0.99 [0.97, 1.11]	n/a	1.016 [1.00, 1.05]

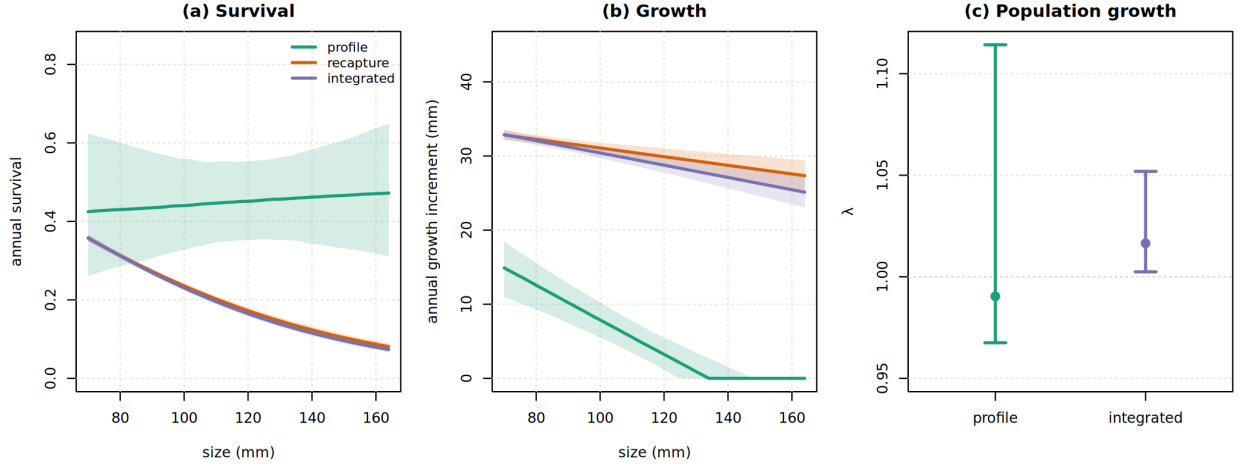


Figure 10: West Brook brook trout: (a) survival, (b) growth increment, and (c) population growth λ under profile-only, mark-recapture, and integrated analyses (medians with 95% bands/intervals). The profile-only fit underestimates growth and misses the survival size gradient that the individual data reveal.

► [COAUTHOR INPUT NEEDED, biology/ecology]: *Biological interpretation of these contrasts: is the $\sim 32 \text{ mm yr}^{-1}$ recapture growth consistent with known brook trout growth in this system? Is the declining apparent survival with size attributable to emigration of larger fish, size-selective mortality, or both? What would the management or ecological reading of the profile-only versus integrated λ be? Please also confirm the framing of the size-dependent-detection caveat for electrofishing.*

6.3 Profile-only uncertainty is multimodal and concentrates on an iso- λ manifold

Replacing the Laplace approximation with full MCMC (four adaptive-Metropolis chains) exposes the non-identifiability directly. The chains do not converge on the vital rates (split $\hat{R}$ up to 6.6), which is a property of the target rather than a sampler failure. They settle in biologically distinct regimes (survival at 80 mm from 0.005 to 0.12, growth increment from 3 to 14 mm, recruitment over more than an order of magnitude) that all reproduce the near-stationary profiles of Figure 9. Across the pooled posterior, however, the dominant eigenvalue varies by only 2.5% (coefficient of variation), against 222% for survival at 80 mm and 131% for recruitment intensity (Figure 11), as Proposition 2 predicts. The Laplace mode (survival about 0.43) does not lie where the posterior mass concentrates, so its narrow intervals for the confounded directions are not trustworthy. We therefore report λ as the profile-identified functional and treat profile-only vital-rate point estimates as non-identified absent individual data; the integrated analysis both resolves the rates and shifts λ to 1.016 [1.00, 1.05].

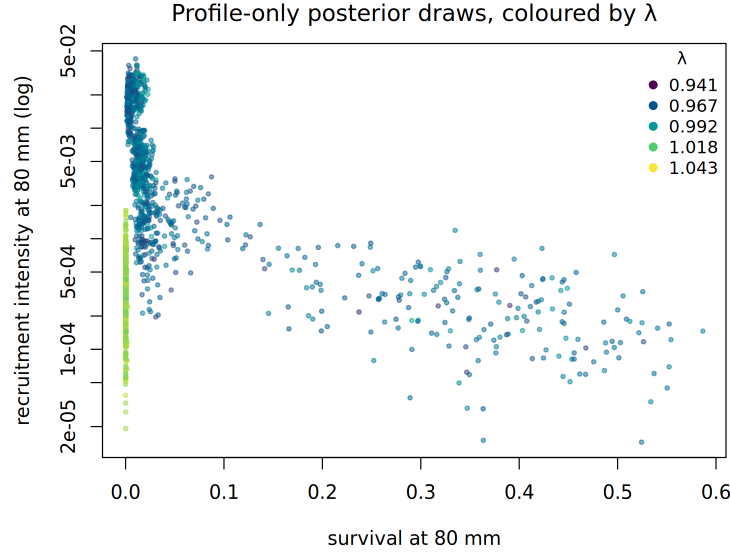


Figure 11: Profile-only posterior draws (West Brook brook trout) in the survival–recruitment plane, coloured by the implied dominant eigenvalue λ . Biologically very different vital-rate combinations span the plane, yet λ varies little (CV 2.5% vs. 222% for survival): profiles identify the asymptotic functional, not the rates that generate it.

This non-identification is invisible to local diagnostics. The observed Fisher information at the Laplace mode spans seven orders of magnitude, and nine of the eleven information eigen-directions contract by more than half from the prior (Figure 12); the single flat direction there is the recruit-size location, unconstrained only because recruitment into the window is near zero at that mode. A Hessian- or Laplace-based identifiability check would therefore declare the model essentially identified, even though the global posterior is the multimodal iso- λ manifold of Figure 11. Local curvature certifies practical identifiability at a single point; establishing global identifiability requires manifold-revealing MCMC, profile likelihoods, or data cloning.

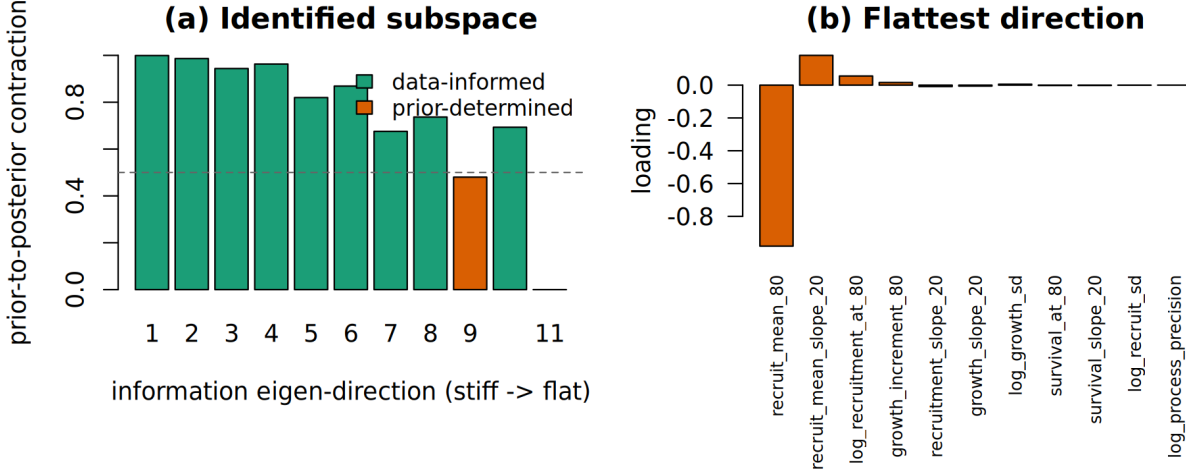


Figure 12: (a) Prior-to-posterior contraction along the observed-information eigen-directions at the West Brook profile-only mode; nine of eleven exceed 0.5, suggesting local identifiability. (b) The flattest direction loads almost entirely on recruit-size location. Local curvature does not reveal the global non-identification of Figure 11.

6.4 Kernel functionals: the population is near its stable distribution

Although the vital rates are not identified, the kernel functionals are. The profile-only posterior gives a well-determined asymptotic growth rate ($\lambda = 0.978 [0.958, 1.034]$) and stable size distribution: the mean per-bin posterior coefficient of variation of the stable distribution is 0.13, roughly seventeen times tighter than that of survival. Both λ and the stable shape are pinned down across the iso- λ manifold, as Proposition 2 predicts. The model-implied stable distribution closely matches the observed average size profile over the bulk of the size range (Figure 13), confirming that the fitted kernel is self-consistent and that the population is observed near its stable distribution, the regime in which the vital rates are non-identified. The one systematic departure, at the smallest observed sizes, is consistent with under-detection of small fish near the lower window edge. This points to the constructive reading of the analysis: from size profiles one should report and trust the kernel functionals (λ and the stable distribution), while the underlying vital rates require individual data.

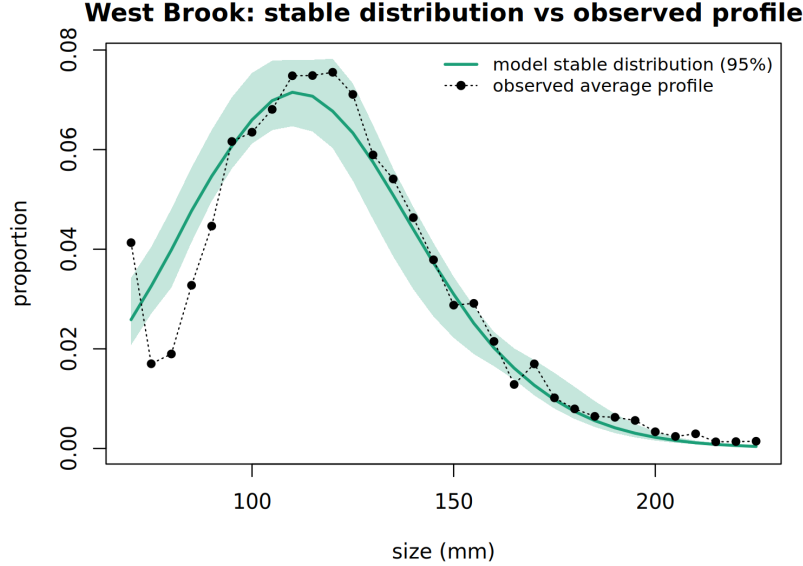


Figure 13: Model-implied stable size distribution (median and 95% band) from the West Brook profile-only posterior, against the observed average size profile. The close match over the bulk of the range shows the population is near its stable distribution; the departure at the smallest sizes is consistent with under-detection near the lower observation window edge.

6.5 External validation against published estimates

The West Brook system has been studied with purpose-built capture–recapture and matrix models (Letcher et al., 2007, 2015; Kanno et al., 2014), giving an external benchmark independent of our analysis (Table 3). Two differences must be reconciled. The first is time step: the published mean seasonal survival of 0.76 (Letcher et al., 2015), compounded over four seasons, implies an annual survival near 0.33, close to our annual apparent survival of 0.28, and the published monthly survivals of 0.89–0.94 (Letcher et al., 2007) bracket the same range. The second, and more important, is that the published models separate permanent emigration from mortality using boundary antennas, so their “survival” is close to true survival, whereas our mark–recapture estimate is apparent survival that confounds death with emigration. Ours is therefore expected to be modestly lower, by roughly the emigration rate (lifetime mainstem–tributary movement of 28–33%; Kanno et al., 2014). Our finding that apparent survival declines with size matches the consistently negative size effect on seasonal survival in Letcher et al. (2015), plausibly amplified because larger fish both die and emigrate more. Our integrated growth (about 32 mm yr^{-1} at 80 mm) is consistent with the spring-dominated seasonal increments of Letcher et al. (2015); our annual detection (0.97) is what four seasonal occasions at the published per-occasion $p = 0.57$ would produce; and our integrated $\lambda = 1.016$ is close to the only published estimate, $\lambda = 1.0008$ (Letcher et al., 2007). The profile-plus-recapture analysis thus recovers the externally validated demography of this population.

Table 3: External validation of the West Brook estimates against the published literature. Published survival is true survival (emigration modelled separately); ours is apparent survival, expected to be modestly lower.

Quantity	This study	Published (West Brook)
Annual survival	0.28 (apparent)	≈ 0.33 from seasonal 0.76 (Letcher et al., 2015); monthly 0.89–0.94 (Letcher et al., 2007)
Survival vs. size	declining, 0.31→0.12	negative size effect (Letcher et al., 2015)
Growth incr. @ 80 mm	32 mm yr ⁻¹	14 mm/season, spring-dominated (Letcher et al., 2015)
Detection	0.97 (annual)	0.57 per occasion $\Rightarrow$ 0.97 over 4 (Letcher et al., 2015)
λ	1.016 [1.00, 1.05]	1.0008 [0.990, 1.012] (Letcher et al., 2007)

7 Discussion

Repeated size profiles are informative about some vital rates (survival, the growth distribution, and recruit size) and almost uninformative about others, chiefly recruitment intensity and, through it, the asymptotic growth rate. The confounding is structural, a ridge that trades recruitment level against its size gradient. It is sharpened by short series and near-stationary distributions, and it is easily masked by an informative prior or by unmodeled size-dependent detection. The brook trout application shows what is at stake: a profile-only fit that passes every internal check still misstates growth by a factor of two and inverts the survival size pattern, errors visible only against individual data.

For practice we recommend declaring the estimand before fitting; reporting prior-to-posterior contraction and the recruitment level/slope correlation as routine identifiability diagnostics; modeling, or at minimum bounding, size-dependent detection; and expressing external information as an explicit, transportability-checkable study design. Where individual data exist, an integrated analysis is worth far more than its marginal cost, though its intervals must be interpreted cautiously when the profile and individual components disagree.

Limitations include reliance on a parametric vital-rate basis (flexible spline bases are a direct extension), a closed-population deterministic projection likelihood (the finite-population particle-filtered variant is developed separately), and a single applied system. The framework and diagnostics themselves are general to size- and stage-structured inverse problems.

► **[COAUTHOR INPUT NEEDED, biology/ecology]:** *Two to three sentences on the ecological/management implications and on how these recommendations would change analyses in your systems, plus any domain-specific limitations we should acknowledge.*

Data and Code Availability

The West Brook data are public domain (CC0) and available from the U.S. Geological Survey (U.S. Geological Survey, 2024); a fetch script is provided. All simulation code, analysis scripts, and figure-generation code are in the project repository.

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► [COAUTHOR INPUT NEEDED, biology/ecology]: *Funding sources, field crews, and data-collection acknowledgments for the West Brook program.*

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